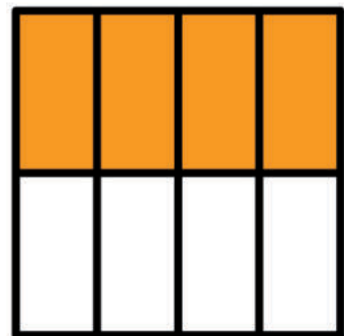
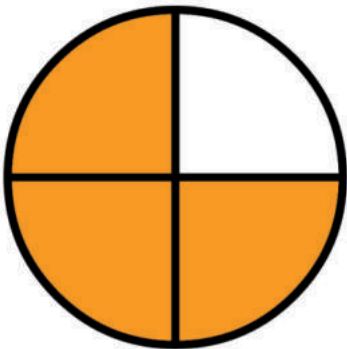


6

FRACTIONS





Definition:

A fraction is a mathematical expression representing a part of a whole or a ratio between two numbers.

Types of fraction:

- **Proper Fraction:**

The numerator is smaller than the denominator.

Such as $\frac{3}{8}$, $\frac{1}{5}$

- **Improper Fraction:**

The numerator is greater than the denominator.

Such as $\frac{8}{5}$, $\frac{9}{2}$

- **Equivalent Fraction:**

Equivalent fractions represent the same value or the same part of a whole, even though they may look different.

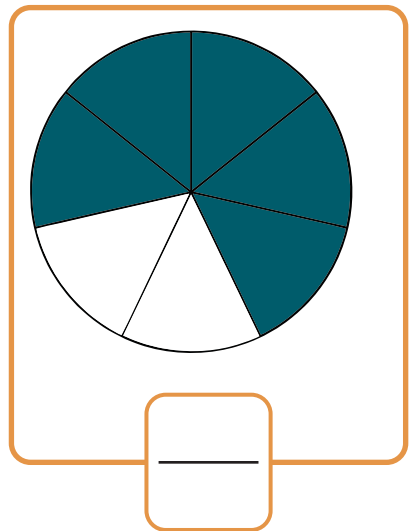
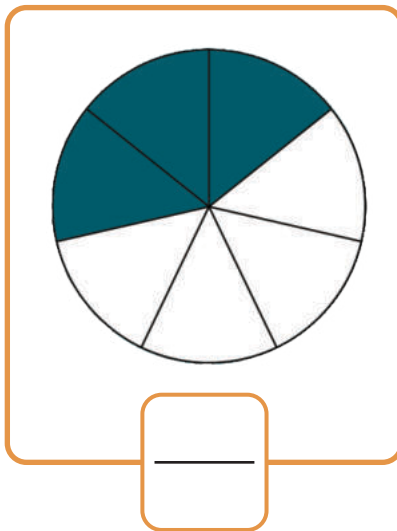
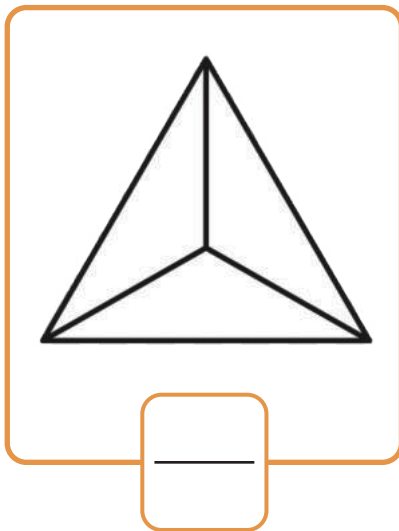
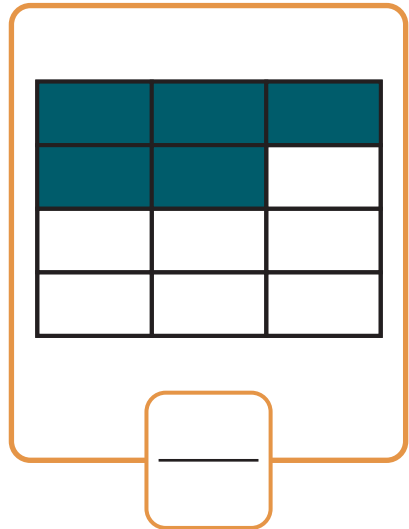
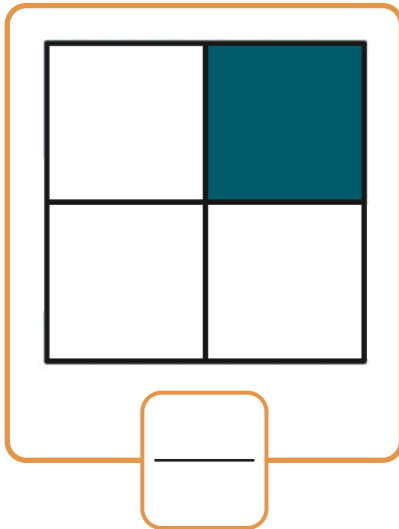
Such as $\frac{1}{2}$ and $\frac{4}{8}$



FRACTIONS

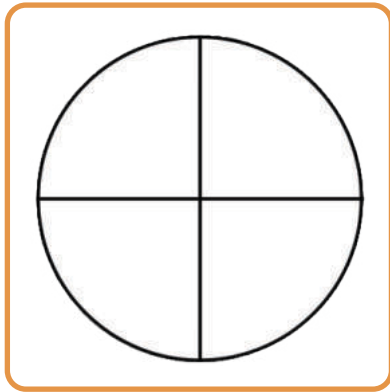


- Express the fractions of coloured parts of the given figures numerically.

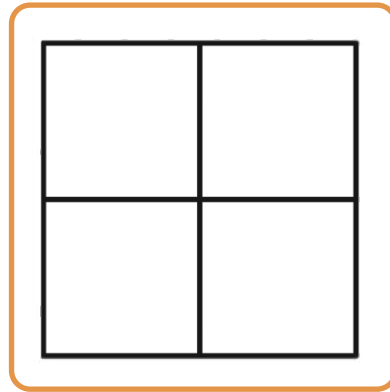




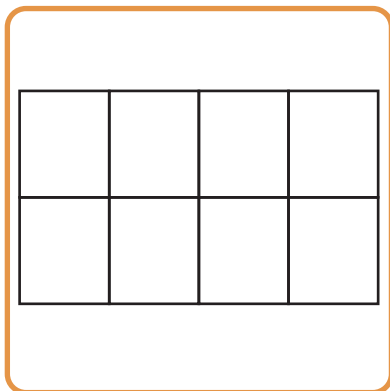
- **Colour the following figures to express the given fractions.**



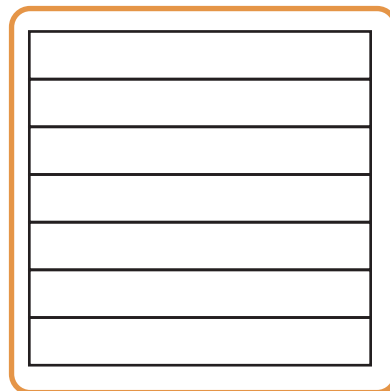
$$\frac{1}{4}$$



$$\frac{3}{4}$$



$$\frac{5}{8}$$



$$\frac{6}{7}$$

- **Write the fractions expressed by:**

(1) Numerator = 4, Denominator = 11 , fraction =

(2) Numerator = 7, Denominator = 8 , fraction =

- Write proper and improper fractions in the following.

1. $\frac{4}{8} \rightarrow$ _____

2. $\frac{63}{65} \rightarrow$ _____

3. $\frac{21}{5} \rightarrow$ _____

4. $\frac{8}{43} \rightarrow$ _____

5. $\frac{50}{51} \rightarrow$ _____

6. $\frac{33}{13} \rightarrow$ _____

7. $\frac{9}{13} \rightarrow$ _____

8. $\frac{7}{2} \rightarrow$ _____

9. $\frac{10}{100} \rightarrow$ _____

10. $\frac{24}{15} \rightarrow$ _____

11. $\frac{8}{7} \rightarrow$ _____

12. $\frac{18}{18} \rightarrow$ _____

13. $\frac{11}{50} \rightarrow$ _____

14. $\frac{9}{4} \rightarrow$ _____

15. $\frac{43}{71} \rightarrow$ _____

16. $\frac{3}{82} \rightarrow$ _____

17. $\frac{2}{22} \rightarrow$ _____

18. $\frac{19}{15} \rightarrow$ _____



- Sort the following fractions into "Proper Fractions" and "Improper Fractions"

$\frac{2}{9}$	$\frac{4}{3}$	$\frac{11}{7}$	$\frac{6}{1}$	$\frac{2}{5}$	$\frac{7}{9}$	$\frac{8}{1}$	$\frac{1}{3}$	$\frac{14}{13}$	$\frac{12}{15}$
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Proper Fractions	Improper Fractions

- Identify and write the figures which show equivalent fractions.

i) $\frac{2}{4} = \frac{\quad}{8}$

ii) $\frac{3}{6} = \frac{\quad}{2}$

iii) $\frac{1}{2} = \frac{\quad}{4}$

iv) $\frac{4}{8} = \frac{\quad}{32}$

v) $\frac{2}{5} = \frac{\quad}{25}$

vi) $\frac{2}{4} = \frac{\quad}{16}$

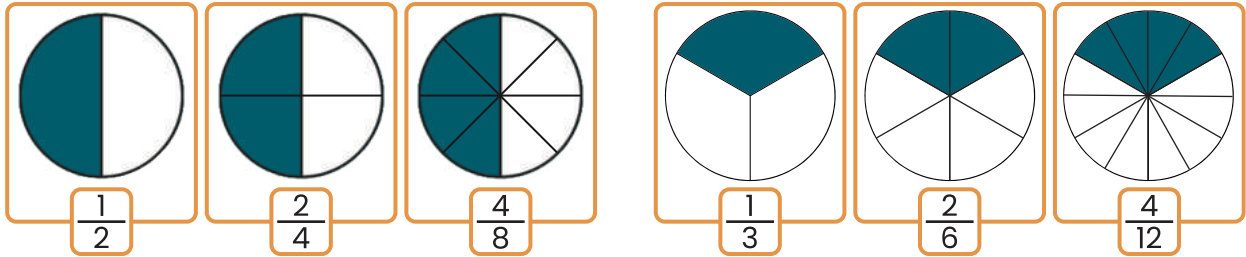
vii) $\frac{2}{9} = \frac{\quad}{81}$

viii) $\frac{4}{7} = \frac{\quad}{49}$

ix) $\frac{5}{10} = \frac{\quad}{100}$

x) $\frac{4}{6} = \frac{\quad}{36}$

- Fill in the missing numbers to make equivalent fractions.



$$\frac{2}{6} = \frac{\quad}{3} \quad \frac{1}{2} = \frac{2}{\quad} \quad \frac{1}{1} = \frac{3}{\quad}$$

$$\frac{3}{4} = \frac{\quad}{8} \quad \frac{2}{4} = \frac{\quad}{16} \quad \frac{1}{2} = \frac{5}{\quad}$$

$$\frac{4}{4} = \frac{\quad}{8} \quad \frac{1}{6} = \frac{2}{\quad} \quad \frac{1}{2} = \frac{\quad}{6}$$

$$\frac{2}{6} = \frac{\quad}{18} \quad \frac{1}{3} = \frac{\quad}{9} \quad \frac{1}{5} = \frac{\quad}{15}$$



FRACTIONS



- **Equivalent fractions based on fraction circles ($\frac{1}{2}$, $\frac{2}{4}$, $\frac{4}{8}$, etc.)**

$$\frac{1}{2} =$$

$$\frac{\quad}{4}$$

$$\frac{1}{3} =$$

$$\frac{\quad}{6}$$

$$\frac{2}{6} =$$

$$\frac{\quad}{12}$$

$$\frac{1}{2} =$$

$$\frac{\quad}{8}$$

$$\frac{1}{3} =$$

$$\frac{\quad}{12}$$

$$\frac{2}{6} =$$

$$\frac{\quad}{3}$$

$$\frac{2}{4} =$$

$$\frac{\quad}{8}$$

$$\frac{4}{8} =$$

$$\frac{\quad}{2}$$

$$\frac{4}{12} =$$

$$\frac{\quad}{3}$$

$$\frac{2}{4} =$$

$$\frac{\quad}{2}$$

$$\frac{4}{8} =$$

$$\frac{\quad}{4}$$

$$\frac{4}{12} =$$

$$\frac{\quad}{6}$$

- Find the sum of following.

$$\frac{1}{9} + \frac{4}{9} = \underline{\hspace{2cm}}$$

$$\frac{3}{6} + \frac{2}{6} = \underline{\hspace{2cm}}$$

$$\frac{6}{8} + \frac{1}{8} = \underline{\hspace{2cm}}$$

$$\frac{4}{7} + \frac{2}{7} = \underline{\hspace{2cm}}$$

$$\frac{5}{7} + \frac{1}{7} = \underline{\hspace{2cm}}$$

$$\frac{1}{5} + \frac{1}{5} = \underline{\hspace{2cm}}$$

$$\frac{3}{5} + \frac{1}{5} = \underline{\hspace{2cm}}$$

$$\frac{2}{9} + \frac{3}{9} = \underline{\hspace{2cm}}$$

$$\frac{2}{8} + \frac{1}{8} = \underline{\hspace{2cm}}$$

$$\frac{2}{4} + \frac{1}{4} = \underline{\hspace{2cm}}$$

$$\frac{1}{5} + \frac{3}{5} = \underline{\hspace{2cm}}$$

$$\frac{6}{8} + \frac{1}{8} = \underline{\hspace{2cm}}$$

$$\frac{4}{5} + \frac{4}{5} = \underline{\hspace{2cm}}$$

$$\frac{1}{6} + \frac{3}{6} = \underline{\hspace{2cm}}$$

$$\frac{4}{8} + \frac{2}{8} = \underline{\hspace{2cm}}$$

$$\frac{3}{7} + \frac{2}{7} = \underline{\hspace{2cm}}$$

$$\frac{1}{10} + \frac{6}{10} = \underline{\hspace{2cm}}$$

$$\frac{5}{8} + \frac{1}{8} = \underline{\hspace{2cm}}$$

$$\frac{7}{9} + \frac{1}{9} = \underline{\hspace{2cm}}$$

$$\frac{2}{4} + \frac{1}{4} = \underline{\hspace{2cm}}$$



FRACTIONS



- Subtract the following.

$$\frac{7}{9} - \frac{1}{9} = \underline{\hspace{2cm}}$$

$$\frac{4}{6} - \frac{2}{6} = \underline{\hspace{2cm}}$$

$$\frac{6}{9} - \frac{5}{9} = \underline{\hspace{2cm}}$$

$$\frac{5}{7} - \frac{4}{7} = \underline{\hspace{2cm}}$$

$$\frac{3}{6} - \frac{0}{6} = \underline{\hspace{2cm}}$$

$$\frac{5}{8} - \frac{1}{8} = \underline{\hspace{2cm}}$$

$$\frac{5}{9} - \frac{4}{9} = \underline{\hspace{2cm}}$$

$$\frac{2}{5} - \frac{1}{5} = \underline{\hspace{2cm}}$$

$$\frac{4}{9} - \frac{3}{9} = \underline{\hspace{2cm}}$$

$$\frac{5}{8} - \frac{2}{8} = \underline{\hspace{2cm}}$$

$$\frac{5}{3} - \frac{3}{3} = \underline{\hspace{2cm}}$$

$$\frac{8}{5} - \frac{2}{5} = \underline{\hspace{2cm}}$$

$$\frac{12}{7} - \frac{3}{7} = \underline{\hspace{2cm}}$$

$$\frac{13}{6} - \frac{5}{6} = \underline{\hspace{2cm}}$$

$$\frac{17}{8} - \frac{3}{8} = \underline{\hspace{2cm}}$$

$$\frac{23}{10} - \frac{3}{10} = \underline{\hspace{2cm}}$$

$$\frac{27}{11} - \frac{4}{11} = \underline{\hspace{2cm}}$$

$$\frac{31}{9} - \frac{7}{9} = \underline{\hspace{2cm}}$$

$$\frac{27}{15} - \frac{22}{15} = \underline{\hspace{2cm}}$$

$$\frac{23}{5} - \frac{7}{5} = \underline{\hspace{2cm}}$$